Development of a Multi-Branch Coffee Shop Management Web Application Using Spring Boot and REST Architecture

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1. Research of the Problem

1.1 Problem Statement

The rapid digital transformation of the service industry has significantly changed customer expectations and operational standards in the food and beverage sector. Coffee shops, particularly those operating multiple branches, face increasing pressure to digitalize their business processes in order to remain competitive.

Despite this transformation, many small and medium-sized coffee chains still rely on fragmented systems for managing reservations, menus, and customer feedback. Common issues include:

- Manual reservation management via phone calls or social media platforms
- Double-booking or overlapping reservations
- Lack of centralized data management across multiple branches
- Absence of analytical tools for business insights
- Limited digital interaction between customers and the coffee shop
- No integration of QR-based services
- Inefficient review moderation processes

These limitations result in operational inefficiencies, reduced customer satisfaction, and missed opportunities for data-driven decision-making.

1.2 Context and Significance

The hospitality industry increasingly depends on digital infrastructure to optimize operations and improve customer experience. According to industry reports, online reservation systems improve table utilization rates and reduce administrative workload. Furthermore, the adoption of QR-based digital menus accelerated globally after the COVID-19 pandemic, becoming a standard feature in modern restaurants and cafés.

For multi-branch coffee shops, the challenge is more complex. They require centralized management while maintaining branch-level autonomy. A scalable, secure, and modular system is necessary to ensure consistency across locations while allowing localized control.

Therefore, the identified problem is the absence of a production-ready, customizable, and cost-effective web platform tailored specifically for multi-branch coffee shops that integrates reservation management, digital menus, QR functionality, analytics, and administrative control within a unified REST-based architecture.

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2. Literature Review and Comparison

To better understand existing solutions and identify research gaps, several commercial restaurant management platforms were analyzed.

2.1 OpenTable

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OpenTable is a widely used restaurant reservation platform that provides table management, booking automation, and customer engagement tools.

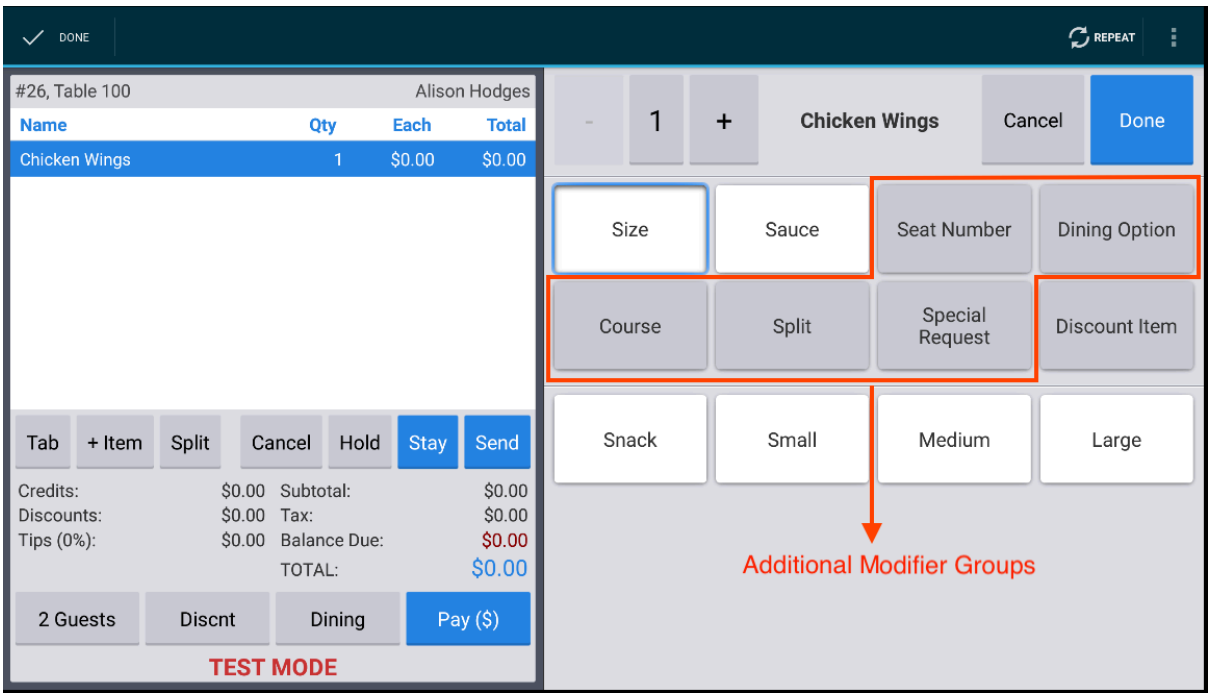
Strengths:

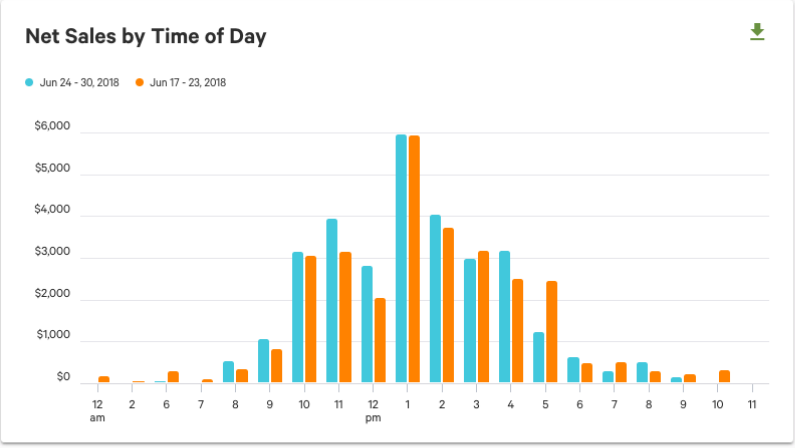
- Advanced reservation management
- Automated booking confirmations
- Business analytics tools
- Large customer network

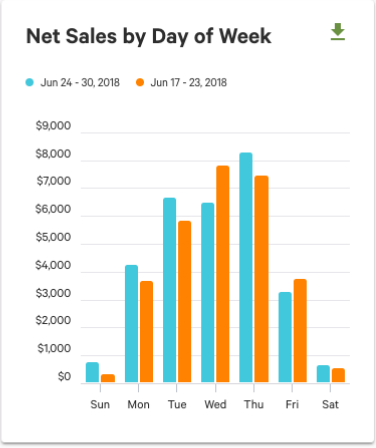
Weaknesses:

- High subscription costs
- Limited customization for internal business logic
- Dependence on third-party infrastructure
- Not optimized for small local coffee chains with specific operational requirements

2.2 Toast POS







Payment Summary

Expand All | Collapse All

TYPE	TRANSACTIONS	AMOUNT	NON-CASH TIPS	TOTAL
Cash	1268	$33,299.70	$0.00	$33,299.70
Credit	556	$16,750.24	$3,894.53	$20,644.77
Gift Card	141	$2,195.30	$340.21	$2,535.51
Other	17	$399.67	$10.78	$410.45
House Account	8	$100.83	$13.86	$114.69

Top 5 Sales Categories

1 Food $25,708.32	1 Food $21,974.73
2 Alcohol $2,784.03	2 No Category $3,982.82
3 No Category $1,223.91	3 Alcohol $2,884.04
4 NA Bev $118.20	4 NA Bev $159.75
5 James Test $108.58	5 James Test $4.22

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Toast is a comprehensive restaurant management and POS system integrating payments, reservations, and analytics.

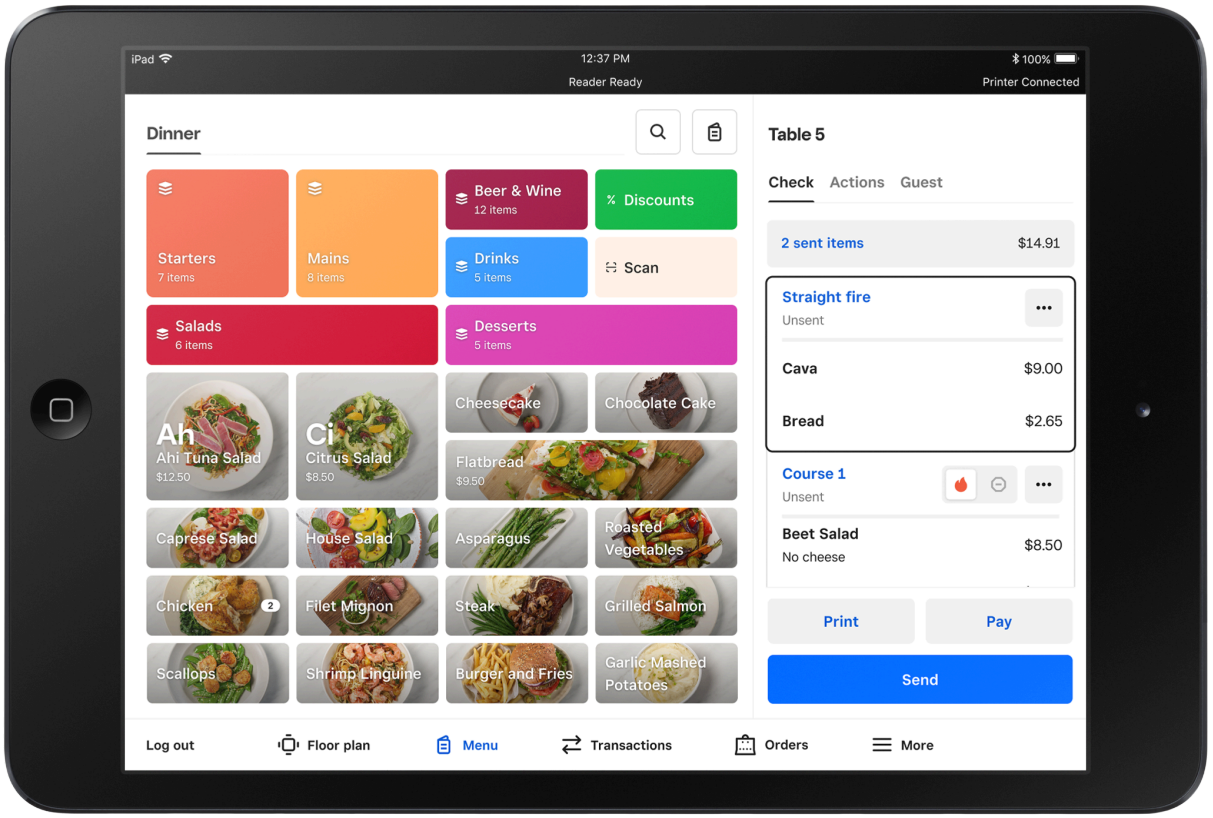
Strengths:

- Integrated POS and analytics
- Payment processing
- Inventory management

Weaknesses:

- Oriented primarily toward medium and large restaurants
- High implementation and hardware costs
- Complex system architecture for small businesses
- Limited flexibility in modifying core logic

2.3 Square for Restaurants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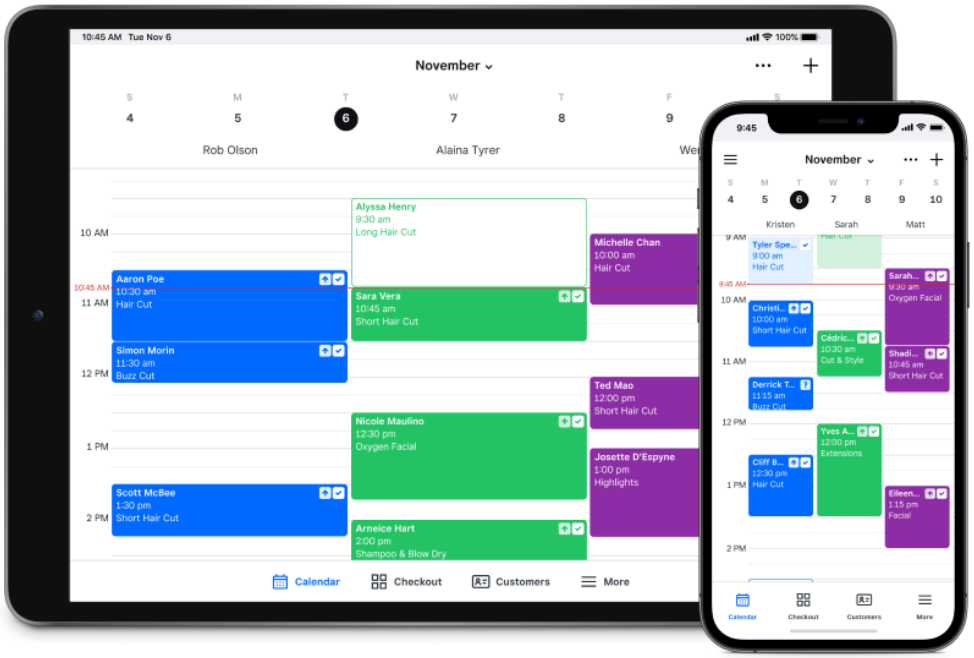
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Aaron Poe
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Simon Morin
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Sara Vera
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Nicole Maulino
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Michelle Chan
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Ted Mao
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Short Hair Cut
Josette D'Espyne
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Highlights
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Kristen
Sarah
Matt

Square provides digital payment solutions and restaurant management tools.

Strengths:

- User-friendly interface
- Integrated payment ecosystem
- Basic reservation features

Weaknesses:

- Not designed for multi-branch coffee chains with centralized control
- Restricted backend customization
- Limited branch-level role hierarchy management

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3. Analysis and Summary

The comparative analysis of existing systems reveals several critical observations:

1. Most available solutions operate as Software-as-a-Service (SaaS) platforms, limiting direct control over business data.
2. Customization of reservation algorithms and role-based access logic is restricted.
3. Multi-branch management is often treated as an add-on rather than a core architectural feature.
4. QR-based integration (menu, reservation, feedback, promotions) is not unified within a single ecosystem.
5. Subscription and transaction fees may be financially burdensome for small coffee chains.

Identified Research Gap

There is a lack of a lightweight, production-ready, REST-based web application specifically designed for multi-branch coffee shops that:

- Provides centralized control with branch-level autonomy
- Implements a customizable anti-overlapping reservation algorithm
- Integrates QR-based services into a single ecosystem
- Supports role-based access control with multiple administrative levels
- Ensures full ownership and control of data
- Is scalable and deployable using modern DevOps practices (Docker, CI/CD)

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4. Proposed Solution and Value Proposition

4.1 Proposed Solution

This project proposes the development of a full-stack web application for multi-branch coffee shop management using:

- Java Spring Boot for backend development
- Spring Security with JWT and OAuth2 for authentication
- PostgreSQL as the relational database
- RESTful API architecture
- React for frontend implementation
- Docker for containerization
- Swagger (OpenAPI) for API documentation
- CI/CD pipeline for deployment automation

4.2 Core Functional Modules

1. Authentication Module
    - Email-based registration
    - Google OAuth2 login
    - JWT-based stateless authentication
    - Role-based authorization (User, Admin, SuperAdmin)
2. Multi-Branch Management
    - Branch creation and configuration
    - Branch-specific menu and tables
    - Centralized control by SuperAdmin
3. Reservation Engine
    - Prevention of time-slot overlaps
    - Editable and cancellable reservations
    - Administrator confirmation workflow
    - Capacity-aware booking logic
4. Menu and Category Management
    - Category-based structure (Coffee, Desserts, Breakfast)
    - Image upload and storage
    - Discount system
    - Availability status
5. Review System
    - Registered users only
    - 1–5 star rating system
    - Moderation before publication
6. Analytics Module
    - Daily reservation statistics
    - Branch performance metrics
    - Graph-based visualizations
7. QR Ecosystem Module
    - QR for each table (direct branch menu access)
    - QR for branch-specific landing pages
    - QR for promotional campaigns
    - QR for quick review submission

4.3 Value Proposition

For Coffee Shop Owners:

- Full control over business data
- Elimination of reservation conflicts
- Centralized management of multiple branches
- Cost efficiency compared to subscription-based SaaS platforms
- Scalable and secure architecture

For Customers:

- Convenient online reservations
- Transparent menu with pricing and availability
- Faster access through QR codes
- Reliable booking confirmation

For Technical Sustainability:

- Production-ready architecture
- Microservice-ready structure for future scaling
- Cloud-deployment compatibility
- Extensible modular design

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5. Conclusion

The conducted research demonstrates that while commercial restaurant management systems provide robust functionality, they lack flexibility, affordability, and architectural transparency for small and medium-sized multi-branch coffee shops.

This project addresses the identified research gap by proposing a production-ready, REST-based, multi-branch management platform built with Spring Boot and PostgreSQL. The integration of a QR ecosystem, role-based architecture, and customizable reservation algorithms creates a comprehensive digital solution tailored specifically for coffee shop chains.

The system is designed not only as a practical business tool but also as a scalable technological foundation that aligns with modern software engineering standards.